

ZeroPad1d#

<https://pytorch.org/docs/stable/generated/torch.nn.ZeroPad1d.html#torch.nn.Z>

Description:

Pads the input tensor boundaries with zero. For N-dimensional padding, use `torch.nn.functional.pad()`. `padding` (int, tuple) – the size of the padding. If is int, uses the same padding in both boundaries. If a 2-tuple, uses $(padding_left, padding_right)$. Input: (C, W_{in}) or (N, C, W_{in}) . Output: (C, W_{out}) or (N, C, W_{out}) , where

Function Signature

```
class torch.nn.ZeroPad1d(padding)[source]#
```

Parameters

Shape:

```
extra_repr()[source]#
```

Return type

Returns:

str

Code Examples

```
>>> m = nn.ZeroPad1d(2)
>>> input = torch.randn(1, 2, 4)
>>> input
tensor([[[[-1.0491, -0.7152, -0.0749,  0.8530],
          [-1.3287,  1.8966,  0.1466, -0.2771]]]])
>>> m(input)
tensor([[[[ 0.0000,  0.0000, -1.0491, -0.7152, -0.0749,  0.8530,
            0.0000,
            0.0000],
          [ 0.0000,  0.0000, -1.3287,  1.8966,  0.1466, -0.2771,
            0.0000,
            0.0000]]]])
>>> m = nn.ZeroPad1d(2)
>>> input = torch.randn(1, 2, 3)
>>> input
tensor([[[[ 1.6616,  1.4523, -1.1255],
          [-3.6372,  0.1182, -1.8652]]]])
>>> m(input)
tensor([[[[ 0.0000,  0.0000,  1.6616,  1.4523, -1.1255,  0.0000,
            0.0000],
          [ 0.0000,  0.0000, -3.6372,  0.1182, -1.8652,  0.0000,
            0.0000]]]])
>>> # using different paddings for different sides
>>> m = nn.ZeroPad1d((3, 1))
>>> m(input)
tensor([[[[ 0.0000,  0.0000,  0.0000,  1.6616,  1.4523, -1.1255,
            0.0000],
          [ 0.0000,  0.0000,  0.0000, -3.6372,  0.1182, -1.8652,
            0.0000]]]])
```

Detailed Documentation

Documentation

Pads the input tensor boundaries with zero.

For N-dimensional padding, use `torch.nn.functional.pad()`.

`padding` (int, tuple) – the size of the padding. If is int, uses the same padding in both boundaries. If a 2-tuple, uses $(padding_left, padding_right)$

Input: (C, W_{in}) or (N, C, W_{in})

Output: (C, W_{out}) or (N, C, W_{out}) , where

$$W_{out} = W_{in} + padding_left + padding_right$$

Return the extra representation of the module.